THE LAWS OF SOFTWARE TEAMS

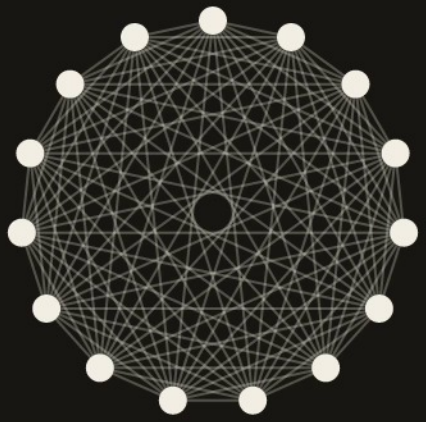
Dunbar's number

People can keep about 150 stable relationships

swipe →

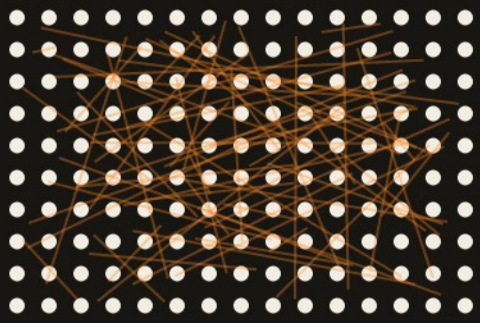
Ivan Rykov

At 15, everyone knows
At 150, nobody does



105

links · 15 people



11,175

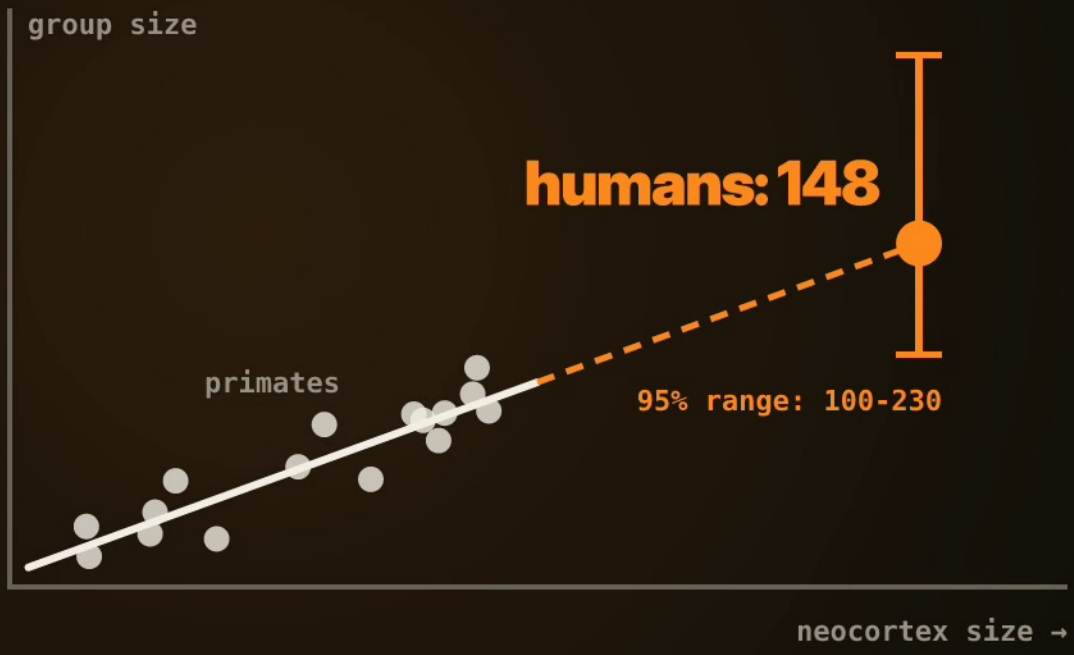
links · 150 people

`links between people = n × (n − 1) / 2`

There is a number for this

Dunbar's number: about 150

Robin Dunbar · anthropologist · 1992-93



Rounded to 150 stable relationships

Dunbar, 1992 and 1993 · primate points illustrative

Friends come in layers



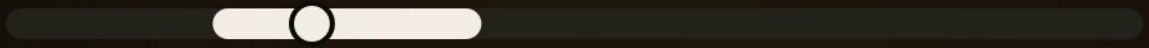
5 close · 15 best friends · 50 good friends · 150 friends

about ×3 per layer · then 500 and 1,500

Dunbar's layers · seen again in phone-call data (MacCarron et al., 2016)

A rough idea, not a law of nature

Dunbar, 1993 — 148 · range 100-230



Re-analysis, 2021 — 69 · range 4-292



other models: averages 16-109 · widest range 5-520

0 100 200 300 400 500

LINDENFORS, WARTEL & LIND · BIOLOGY LETTERS · 2021

"A cognitive limit on human group size cannot be derived in this manner"

Companies use it as a guide

W. L. GORE & ASSOCIATES

A new plant when one grew past about 150-200 people

Gladwell, The Tipping Point (2000): 150 · company histories: about 200

SPOTIFY · 2012

“Tribes are designed to be smaller than 100 people or so”

Kniberg & Ivarsson, Scaling Agile @ Spotify

Each layer needs a new way to talk



up to 10 people

a team · works by talking



~50 people

a team of teams · needs shared rituals

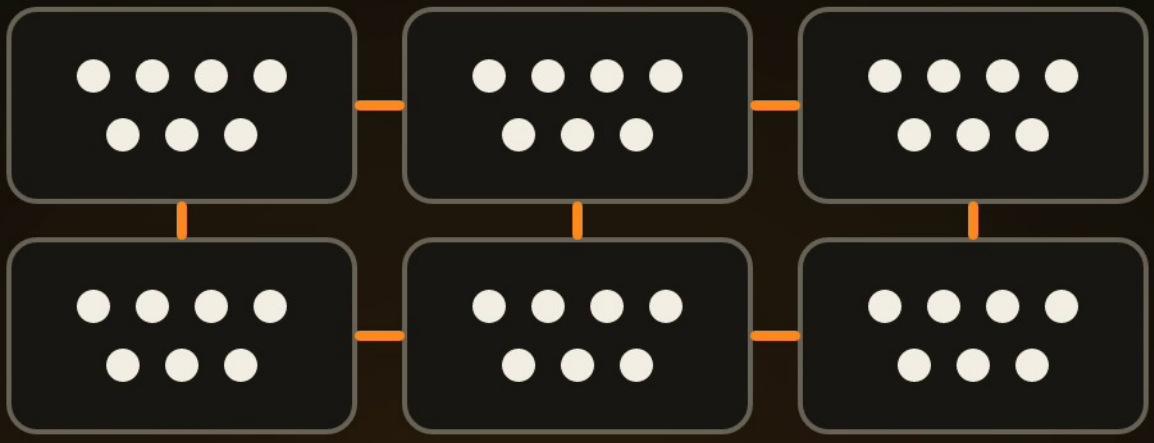


~150 people

a department · needs written decisions

a manager's reading of the layers, not Dunbar's research

Split early, along the system



Before the group stops working

CONWAY'S LAW

Team lines become system lines

illustration · amber: the interfaces between teams

08 TAKEAWAY

Grow in layers, not in one big room



Your org has a size limit, even if nobody wrote it down

HOW TO USE IT

1. Add a new level of structure before the old one stops working. Grow a team only as far as the work needs. I have seen teams grow far beyond that, and more people did not replace clear goals. First the goals, then the size of the team
2. Keep groups small. Ideally, about four engineers work on one thing. A full team, with backend, frontend and QA, has up to ten people. A lead who also does technical work can manage up to ten people directly. A people manager can manage twenty at most
3. When a team grows, split the team or the roles: one lead for people and separate tech leads, sometimes one for each area. Choose who joins which team by the priorities of the work, strengths and how people already work together

HOW TO USE IT

4. **Split teams so that they overlap as little as possible. Give shared work, like the architecture or the test setup, to chapters or guilds. Give harder questions to a small council of strong engineers, so not everyone has to join every discussion**
5. **Connect the teams: demos every two weeks, write-ups of what changes in each team and a monthly talk by leadership about direction**
6. **Hold 1:1s with direct reports about every two weeks, and skip-level 1:1s about once a month or less often. Everyone should know they can go to a neutral person above their manager**

HOW TO USE IT

7. **Hire the first core engineers slowly. Often they are people the team has already worked with. Others join only after a careful check of their values and approach. Mix people who drive, people with ideas and people who do the hard daily work. Later, hiring can go faster**

AI agents can take over much of the coding for a small team, so fewer products will probably need teams of 20 to 100 people. But some limits stay. In practice, one person can keep about five to seven parallel streams of work in mind, and someone has to review the generated code and judge it as a product. Build a product around a core of five to ten strong engineers

Sources: Robin Dunbar, 1992 and 1993 · Lindenfors, Wartel and Lind, 2021 · Dunbar's reply, The Conversation, 2021 · Zhou et al., 2005 · MacCarron et al., 2016 · Kniberg and Ivarsson, 2012

next: the Peter principle